Undirected TSP as a Constrained GSTP Variant

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Abstract

Classical formulations of the Traveling Salesman Problem (TSP) search for a 1-dimensional cycle of edges. We propose a novel formulation that constructs a 2-dimensional surface instead, motivated by the geometric intuition that a tour forms the boundary of a surface. This geometric shift maps symmetric TSP to a constrained Group Steiner Tree Problem variant ($cGSTP$), where maximizing the net weight of this tree is equivalent to minimizing the tour length. Based on this structure, we establish a Mixed-Integer Linear Program (MILP) that leverages the Euler characteristic to enforce topological consistency. The formulation is exact when applied to the complete input complex, despite being computationally intractable for all but the smallest instances. In practice, it provides an effective heuristic when restricted to a sparse candidate set such as Delaunay triangulation.

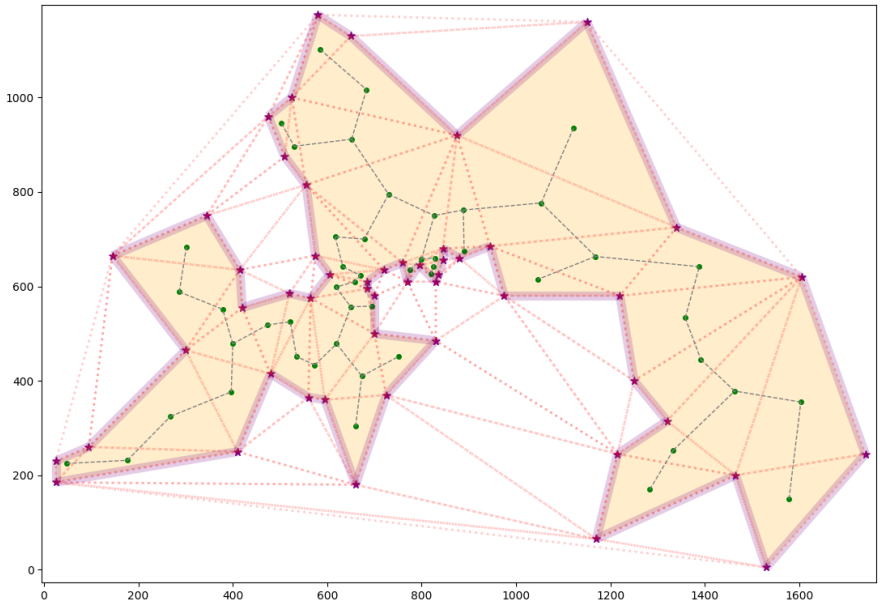

Figure 1 Visual intuition on `berlin52` ($N = 52$): select a connected set of triangles (surface); its boundary is the optimal tour.

1 Introduction

TSP is conventionally modeled on a graph $G = (V, E)$: choose a minimum-weight Hamiltonian cycle by selecting edges $E' \subseteq E$ (Fig. 1). Edge-based models enforce local degree conditions but must also encode global connectivity, often via strong but exponential subtour elimination constraints [7], or via weaker alternatives such as MTZ [22] or flow-based constraints [15]. We propose a shift in perspective from *graph-theoretic* cycle selection to a *topological* construction. Instead of building the tour as a 1-dimensional sequence of edges, we construct a 2-dimensional surface whose boundary is the Hamiltonian cycle. Given a candidate triangle set $\mathcal{T} \subseteq \binom{V}{3}$,

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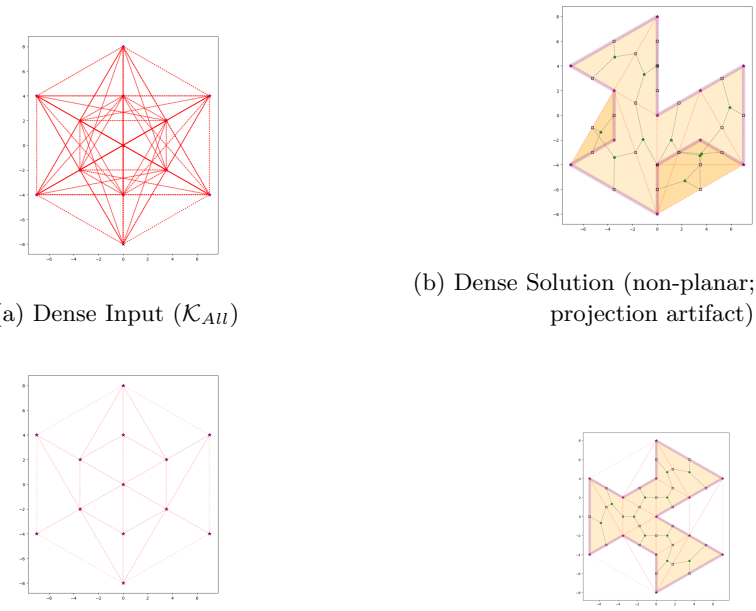
we select a connected subset whose induced boundary length is minimized. When $\mathcal{T}$ is the universal set $\mathcal{K}_{All} := \binom{V}{3}$ (Fig. 2a), the formulation is exact but computationally feasible only for very small N. Restricting $\mathcal{T}$ to a sparse triangulation (e.g., Delaunay, Fig. 2c) yields a highly efficient candidate-set heuristic, though it may render the model infeasible if $\mathcal{T}$ contains no Hamiltonian cycle [10].

Related Work. Classical exact methods rely on branch-and-cut with subtour elimination [1, 7]. Tree-structured connectivity constraints are well-studied in flow-based routing formulations [15] and in Steiner variants [20]. Geometric approximation schemes for Euclidean TSP exploit recursive planar subdivisions [2, 23]. Our formulation aligns with boundary-minimization viewpoints from computational topology [12] and geometry [13, 14], planar duality principles [27], and optimization of cycles on surfaces [5], but differs in that we optimize the *surface itself* via a simplicial incidence structure. This yields a constrained GSTP [25], similar to geometric covering variants like TSP with Neighborhoods [11], whose objective resembles prize-collecting Steiner tree terms [16], except that "prizes" arise from simplicial incidences (i.e., edges of the Steiner tree).



(a) Dense Input ($\mathcal{K}_{All}$)

(b) Dense Solution (non-planar; overlap is a projection artifact)

(c) Delaunay Input ($\mathcal{T}_{Delaunay}$)

(d) Delaunay Solution (Planar)

Figure 2 Nested hexagons ($N = 13$): with all triangles the optimum is an abstract non-planar surface; Delaunay input forces a planar one.

Contribution. Our contribution is fourfold:

(i) We establish (in Sec. 2) an equivalence between min-weight Hamiltonian cycles in the *complete* complex $\mathcal{K}_{All}$ (Fig. 2a) and max-weight dual trees in the bipartite incidence graph (Fig. 3);

(ii) We give a MILP that replaces subtour elimination by a global tree constraint plus a local Euler characteristic filter (Sec. 3);

(iii) We prove (in companion note [3]) that these constraints enforce a disk whose boundary is a single simple Hamiltonian cycle; and

(iv) We demonstrate that the formulation is fully *input-decoupled*: it treats the candidate

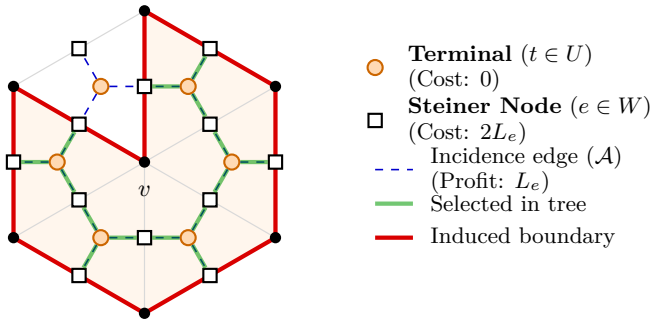


Figure 3 Bipartite incidence graph: triangles x_t (circles) and edges y_e (squares); dashed lines are adjacencies. A feasible solution selects a connected set of triangles (filled orange), inducing the highlighted boundary (red) around the center; translucent green edges indicate incidence edges chosen in the resulting tree.

triangle set $\mathcal{T}$ as a black-box constraint on the search space, working both on $\mathcal{K}_{All} = \binom{V}{3}$ as well as on sparse inputs such as Delaunay or Greedy triangulations [8, 10, 21].

2 Reduction to cGSTP

We first establish the structural transformation of the problem, demonstrating that the TSP on G corresponds to a cGSTP on a dual structure, augmented with topological counting (cardinality and degree) constraints. Given a TSP instance on $G = (V, E)$, we construct the *Bipartite Incidence Graph* $B = (U \cup W, \mathcal{A})$ (see Fig. 3), which represents the connectivity between triangles and edges:

- **Terminals (U):** A candidate triangle set $U := \mathcal{T} \subseteq \binom{V}{3}$.
- **Steiner Nodes (W):** The set of primal edges $W := E$.
- **Adjacency ($\mathcal{A}$):** An edge exists between $t \in U$ and $e \in W$ if triangle t contains edge e.

To establish equivalence, we assign costs and profits as detailed in Table 1, where L_e denotes the metric length of edge e. Each city v (e.g., the central node labeled v in Fig. 3) induces the group

$$U(v) := \{t \in U \mid v \in t\},$$

i.e., the set of candidate triangles incident to v. Group satisfaction is enforced through the group-coverage and topological admissibility constraints.

Table 1 Mapping to a cGSTP: costs, profits, and enforced topological degrees.

Primal Concept	cGSTP Component	Cost & Profit / Degree Constraint
Triangle ($t \in \mathcal{T}$)	Terminal Node ($t \in U$)	Cost 0; Deg $\in \{0, 3\}$ (t selected $\rightarrow$ all edges selected)
Edge ($e \in E$)	Steiner Node ($e \in W$)	Cost $2L_e$; Deg ≤ 2 (manifold condition)
Incidence ($e \subset t$)	Edge in $\mathcal{A}$	Profit L_e (connection reward)
City ($v \in V$)	Group $U(v)$	Target: Visit ≥ 1 terminal in group

Any feasible surface selection induces an active subgraph $B' \subseteq B$. Define its net weight as:

$$NW(B') := \sum_{(t,e) \in \mathcal{A}(B')} profit(t, e) - \sum_{e \in W(B')} cost(e). \tag{1}$$

Internal edges are incident to two selected triangles and contribute $2L_e - 2L_e = 0$ to $NW(B')$; boundary edges are incident to one triangle and contribute $L_e - 2L_e = -L_e$. Hence maximizing $NW(B')$ is equivalent to minimizing the total boundary length (the tour). Section 3 implements this equivalence directly.

3 Mathematical Formulation

We formulate the problem as finding a subgraph in the bipartite incidence graph $B = (U \cup W, \mathcal{A})$ that corresponds to a valid surface. Let $x_t, y_e, z_{te} \in \{0, 1\}$ be decision variables representing the selection of triangles, edges, and their incidences, respectively. The objective is to minimize the length of the boundary $\partial\mathcal{K}$ of the resulting surface complex $\mathcal{K}$:

$$\text{Min } Z = \sum_{e \in W} 2L_e y_e - \sum_{(t,e) \in \mathcal{A}} L_e z_{te} \tag{2}$$

Eq. (2) is exactly $Z = -NW(B')$ for the induced active subgraph B' (Eq. 1), so minimizing Z corresponds to maximizing the net weight.

We enforce topological validity via the following five structural constraints:

1. *Incidence linking (Terminal Node Degrees):* Every active triangle must use all three of its edges:

$$z_{te} \leq y_e$$

$$z_{te} \leq x_t$$

$$\sum_{e \subset t} z_{te} = 3x_t$$

2. *Manifold Condition (Steiner Node Degrees):* Every primal edge must be incident to at most two triangles (Fig. 4A depicts a violation). In the dual, this restricts the degree of Steiner nodes y_e within the incidence graph.

$$y_e \leq \sum_{t \supset e} z_{te} \leq 2y_e$$

3. *Node Cardinality Constraints:* To guarantee $\mathcal{K}$ is a triangulation of a topological disk with N vertices, we enforce $\sum x_t = N - 2$ (triangles) and $\sum y_e = 2N - 3$ (edges).
4. *Global Connectivity (Tree):* B' must form a single connected tree (see [4] for the detailed implementation). This ensures $\mathcal{K}$ is simply connected (topologically a disk).
5. *Local Connectivity (Steiner Groups):* To prevent vertex singularities (e.g., "bowties", Fig. 4B), the neighborhood of each city v must form a simple path (cf. the center vertex in Fig. 3). We enforce this using the local Euler characteristic of the incidence subgraph $H_v \subseteq B'$ [12]:

$$\chi(H_v) = |\mathrm{V}(H_v)| - |E(H_v)| = 1 \quad \forall v \in V. \tag{3}$$

Here H_v denotes the subgraph of B' induced by the triangles and edges incident to v. Under the global tree constraint, enforcing $\chi(H_v) = 1$ leads to a simple path, ruling out disconnected vertex links and local "bowtie" singularities (Fig. 4B). The presence of at least one terminal per group is thus also implicitly enforced [3].

Figure 4 Forbidden anomalies: (A) An edge incident to > 2 triangles; (B) A "bowtie" vertex violating the Euler characteristic filter.

4 Computational Validation and Results

We validated the correctness of the formulation on simple topological cases before measuring performance.

Visual Verification (Topology). To provide a visual verification, we apply the model to an instance consisting of two nested hexagons with a central node ($N = 13$). Fig. 2 compares the solver's behavior under two different candidate triangle sets: the universal set $\mathcal{K}_{All}$ and the Delaunay triangulation $\mathcal{T}_{Delaunay}$. In both cases, the formulation successfully extracts the optimal surface within the chosen input complex.

Validation on Non-Metric Spaces. To test the topological generality of our formulation, we generated random symmetric non-metric instances of small size where edge weights do not satisfy the triangle inequality. The input was $\mathcal{K}_{All}$. The solver successfully extracted valid dual trees and primal tours; Fig. 5 shows a representative instance with $N = 10$. We verified optimality against a standard TSP formulation (MTZ).

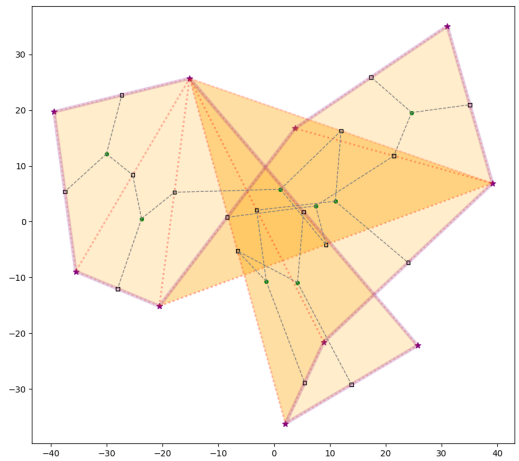


Figure 5 Random symmetric non-metric instance ($N = 10$, visualized by MDS). The topological formulation remains applicable even when the triangle inequality is violated.

Comparison. We restrict comparisons to compact static models, excluding dynamic cut-generation methods (e.g., Concorde [1]). We benchmarked three formulations:

1. **Conventional:** Standard Lifted-MTZ formulation [9] restricted to edges in the triangulation.
2. **SB-MTZ:** Surface-based formulation using MTZ [22]-based tree constraints.
3. **SB-Flow:** Surface-based formulation using Single-Commodity Flow [15]-based tree constraints.

We utilized a planar simplification of the model, which eliminates explicit Steiner node variables y_e, as the incidence graph collapses to the standard dual graph for planar triangulations [4]. For the sake of reproducibility, a MiniZinc [24] implementation is publicly available at

https://github.com/hoketo/surface_based_tsp_with_minizinc

We utilized HiGHS v1.12 [19] on an Apple M4 Max. Table 2 summarizes the performance across all instances, including several from TSPLIB [26]. All experiments use the Delaunay triangulation unless stated otherwise. The results indicate that the SB-Flow generally outperforms SB-MTZ as N grows. For instances ($N \leq 280$), it typically solves at the root. For larger instances, SB-Flow yields a substantially tighter relaxation, reflected in dramatically smaller Branch-and-Bound trees (e.g., 142 nodes vs. 23,205 for `pcb442`).

Table 2 Combined Computational Results. Post-presolve MILP size and performance; values are 10-run means (unless noted), with a timeout limit of 1-hour.

		MILP Size		HiGHS	
Instance	**Model**	**Vars**	**Cons**	**Time (s) [Solved]**	**Nodes**
Berlin52	Lifted-MTZ	341	523	0.08s [10/10]	1
($N = 52$)	SB-MTZ	553	1,107	**0.02s** [10/10]	1
	SB-Flow	684	1,054	0.03s [10/10]	1
St70	Lifted-MTZ	463	723	14.23s [10/10]	2,936
($N = 70$)	SB-MTZ	759	1,521	4.35s [10/10]	257
	SB-Flow	926	1,440	**0.20s** [10/10]	**1**
Ch130	Lifted-MTZ	879	1,362	108.6s [10/10]	16,058
($N = 130$)	SB-MTZ	1,458	2,922	**0.29s** [10/10]	**1**
	SB-Flow	1,793	2,775	1.62s [10/10]	**1**
Tnm199	SB-Flow (Delaunay)	2,764	4,283	**0.51s** [10/10]	**1**
($N = 199$)	SB-Flow (Greedy)	2,090	3,202	4.76s [10/10]	1.6
A280	Lifted-MTZ	1,907	2,992	DNF [0/5]	>53k
($N = 280$)	SB-MTZ	3,188	6,379	29.0s [10/10]	490
	SB-Flow	3,923	6,056	**5.68s** [10/10]	**1**
Pcb442	SB-MTZ	5,195	10,393	1656s* [7/10]	23,205
($N = 442$)	SB-Flow	6,412	9,879	**81.1s** [10/10]	**142**
d657	SB-Flow	9,634	14,837	TLE (0.60%)	>30k
($N = 657$)					

* For `pcb442` SB-MTZ, 3/10 runs exceeded the 1-hour time limit; time and nodes are averaged over the 7 solved runs.

Degenerate Inputs. We tested on `Tnm199` ($N = 199$) from the "Hard-to-Solve" dataset [18]. This instance is constructed to be practically intractable for standard exact solvers; even Concorde requires ≈ 4.7 days to find the global optimum [18]. Our formulation effectively exploits the Greedy triangulation to achieve a high-quality tour with 1.85% gap (vs. 6.17% for Delaunay), surpassing the 5.69% gap of the Christofides heuristic [6]. This demonstrates that the topological formulation scales in quality based on the preferred input complex.

To test the limitations of sparsification under degenerate geometry, we evaluated the instance `p100.100000` ($N = 100$), which arranges points on three parallel lines [17]. The results reveal a critical vulnerability in geometric sparsifiers. While Christofides remains

robust (2.52% gap), Delaunay and Greedy triangulations degrade badly (14%–119% gap), exposing a limitation of sparse candidate sets on near-degenerate geometry.

5 Conclusion

We have presented a surface-based formulation for TSP that shifts the decision space from edges to triangles. This changes the optimization object from a primal cycle to a dual surface: instead of directly enforcing Hamiltonicity on G, we enforce disk-admissibility of a 2-complex whose boundary is the tour. Our experiments indicate that this topological shift can yield improved B&B behavior relative to compact, static edge-based baselines on sparse geometric inputs, particularly when enforcing the global tree constraint via flow. While exact convergence becomes computationally prohibitive for larger instances in our current setup, the results suggest that the surface-based model provides a useful foundation for integrating additional geometric or topological constraints that could be cumbersome in primal edge formulations.

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